

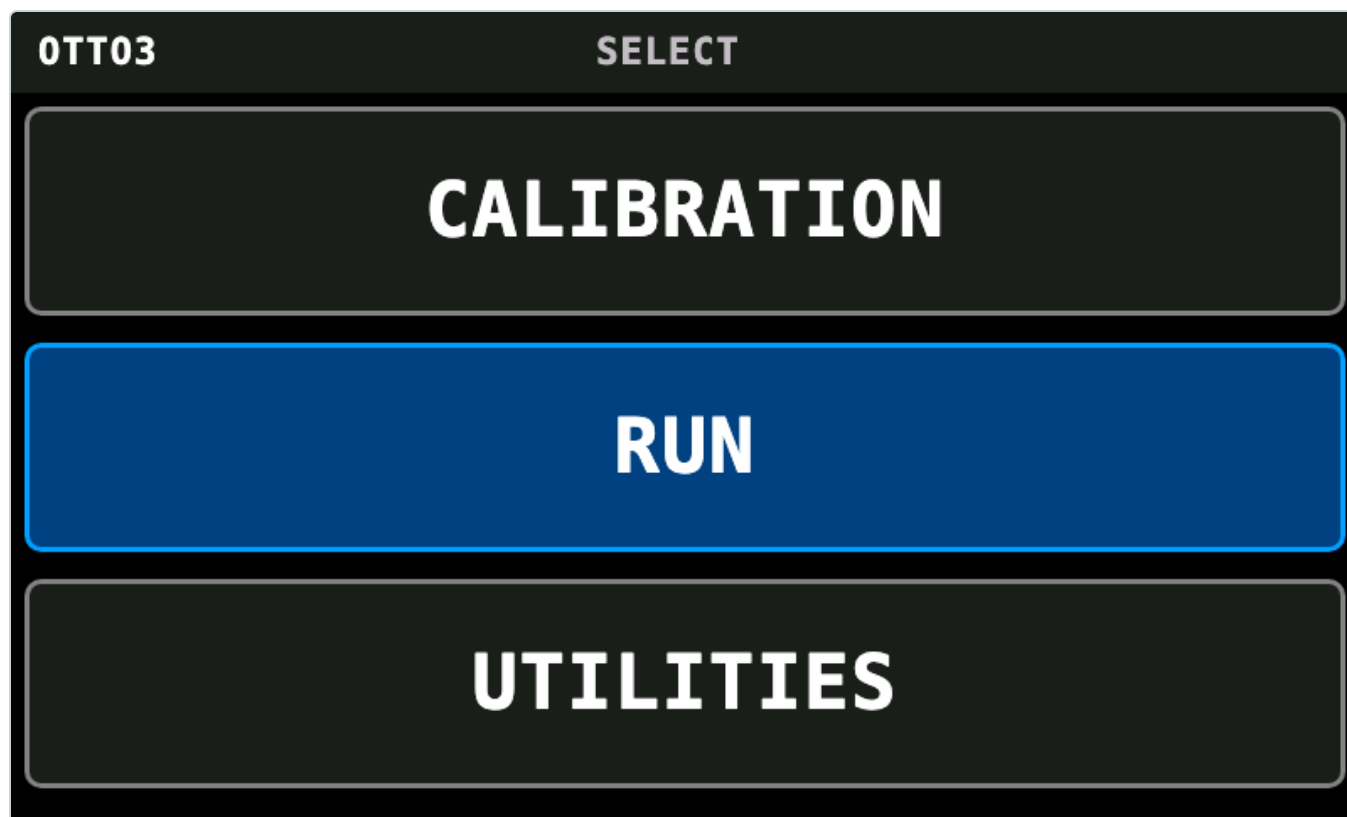
Otto3 operator interface guide

Otto3 is operated through **OttoPanel** — one firmware, two equivalent interfaces: the **touchscreen** on the GIGA Display Shield (no computer needed) and a **remote serial console** (9600 baud, for shell/Claude-driven sessions). Both expose the same actions; the touchscreen adds checklists and confirmation gates, the serial console skips them (the remote operator is the confirmation).

1. Touchscreen

The screens below are renderings generated from the firmware's own layout constants and colors (`OttoPanelUI.h` / `OttoDisplay.h`) — geometry-accurate mockups of the 800×480 shield, not photographs.

Home → **menu** → **checklist** → **run**



- **Home** shows three categories: **CALIBRATION** (steps 1–9), **RUN** (the full sequencing protocol), **UTILITIES** (primes, rinses, dispense/aspirate-all, park, shutdown flush, and the calibration inspection tests).
- Tapping a category opens a **2-column button grid** (cell 1 is always `< BACK`).

CALIBRATION

< BACK

STEP 1 PRIME WASH

STEP 2 PRIME CLEAVAGE

STEP 3 PRIME INCORPORATION

STEP 4 FULL RINSE

STEP 5 DISPENSE CALIBRATION

STEP 6 DISPENSE + ASPIRATE

STEP 7 NESTED CYCLE

STEP 8 ADDREAGENT INC TUNING

STEP 9 ADDREAGENT CLV VALIDATE

- Tapping an action opens its **pre-run checklist**: short physical prerequisites ("RESERVOIRS FULL", "PUMP ARMED (STOP AFTER KEYPAD USE)", "VACUUM OPEN, TRAP IN LINE, 12V ON", ...). Every row must be ticked before **GO** enables (it stays grey until then). **CANCEL** returns to the menu.

CAL: SPLIT TEST
CHECK EVERY BOX, THEN GO

☒ PR2 IN ALL REAGENT SLOTS

☒ MANIFOLD SEATED ON PLATE

☒ VACUUM OPEN, TRAP IN LINE, 12V ON

☐ PORT 6 (AIR) DRY – NO TUBING

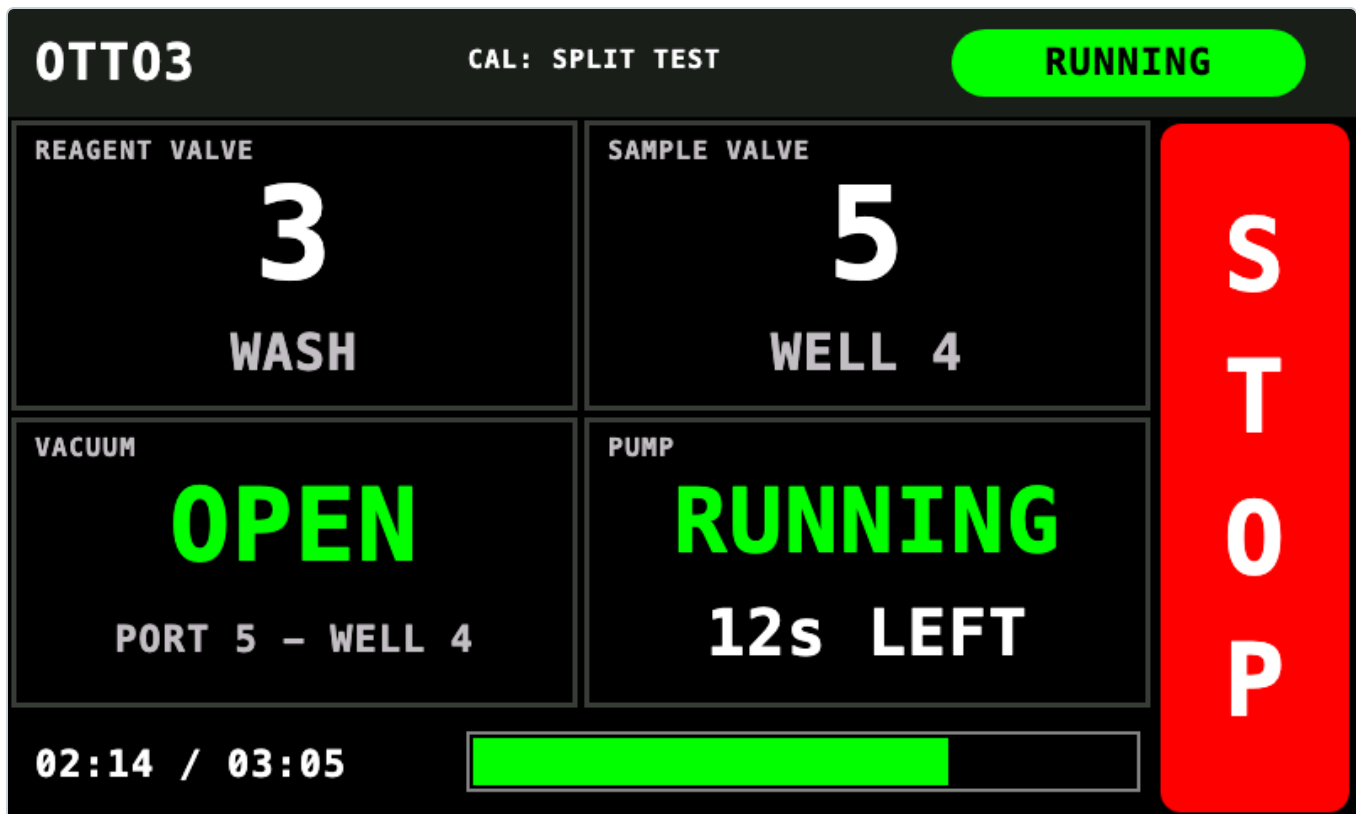
☐ PUMP ARMED (STOP AFTER KEYPAD USE)

EXPECTED ~ 03:05

CANCEL

GO

The run screen



While an action runs, the display is a live dashboard:

- **Four panels** — reagent valve (current port + reagent name), sample valve (current port), vacuum valve + solenoid state, pump state with a live countdown for the current pump segment.
- **Step timer** — elapsed vs expected duration for the whole action (expected times are computed from the live constants at confirm time).
- **Cue banner** — operator instructions timed to the fluidics: calm cues ("PRIMING - NO NEED TO WATCH YET"), armed countdowns ("EYES ON DISP VALVE - FRONT ARRIVES T-11s"), and a triple full-screen **strobe at T-0** so the watch moment is unmissable. During the purge loop each needle gets its own countdown ("RED AT NEEDLE 3").
- **The red STOP column** is always present and always hot. One tap sets the abort latch: the running routine unwinds within ~100 ms (`Wait()` polls the latch in 100 ms slices; pump/valve/solenoid primitives refuse new actions), then the instrument **parks safe** — pump stopped, vacuum solenoid closed, valves to reagent 4 / sample 1 / vacuum 1.

When a cue is armed, the footer becomes the cue banner — amber and pulsing for watch cues, with a live countdown, and a triple full-screen strobe at T-0:

OTT03

CAL: SPLIT TEST

RUNNING

REAGENT VALVE 3 WASH	SAMPLE VALVE 7 WELL 6	STOP
VACUUM CLOSED PORT 8 – PARK	PUMP RUNNING 6s LEFT	
EYES ON DISP VALVE – FRONT ARRIVES T-8s		

The result screen

When an action finishes, a **DONE** screen shows the measured duration and, for calibration tests, an inspection prompt (e.g. "INSPECT: AIR SLUG AT SAME SPOT IN ALL 6 LINES?"). Tap to return to the checklist/menu. (An aborted run shows a red ABORTED screen instead, with the stop time and the parked state.)

CAL: SPLIT TEST

DONE

MEASURED 03:03

INSPECT: AIR SLUG AT SAME SPOT IN ALL 6 LINES?

WELLS ~1 mL EACH

TOUCH SCREEN TO RETURN TO MENU

The Step 5 dispense wizard

CALIBRATION → step 5 runs as a guided wizard: dispense pass → operator enters the measured volume on-screen → the wizard retunes `mLPumpTime` in **RAM** and offers another pass. The converged value must still be written into `OttoFns/constants.ino` by hand (no on-device persistence) — the wizard shows the number to copy.

2. Remote serial console

Open the port at **9600 baud** (or use `tools/otto_console.py`, which does everything below for you). The firmware prints the command table at boot and on `?`. One character per command, newline-terminated.

Command table

Key	Action	Notes
1 – 9	Calibration steps 1–9	checklists skipped
R	FULL RUN (the sequencing protocol)	
w / c / i	Prime wash / cleavage / incorporation, 2 min	
f	FULL RINSE (the flush cycle, ~18 min)	
d	DISPENSE 1 mL to all wells	
a	ASPIRATE all wells	
k	PARK valves (reagent 4 / sample 1 / vacuum 1)	
r	CAL: RESET LINE (wash prime out the vent)	cleans reagent path + vent ONLY
v	CAL: BUBBLE TO VALVE (static)	superseded by <code>b</code> for tuning — static parks read ~0.1 mL low
n	CAL: BUBBLE TO NEEDLE 1 (static)	
t	CAL: SPLIT TEST — full production prefix, stops after the split	
p	CAL: PURGE TEST — the ascending purge only, run right after <code>t</code>	
b	CAL: PARK TEST — <code>t</code> truncated at the park; seals the line so the bubble front freezes for reading	
m	CAL: NEXT SPLIT SLICE — one production slice (descending from 7), reseal after	6× = the full split, stepwise
u	CAL: NUDGE TO 7 — ~13 µL titration step; prints the running total since the park	

Key	Action	Notes
M	CAL: SPLIT ALL 6 — the verbatim production split loop in one step	
x	STOP / ABORT (only accepted during a run)	same latch as the red button
s	STATUS — state, elapsed, live constants	accepted any time
?	Help	

While an action runs the console answers `BUSY (RUNNING)` to everything except `x` and `s`.

Phase markers

Every action streams progress markers, so a script (or Claude) can follow a run without watching the screen:

```

RUN: CAL: SPLIT TEST      <- action accepted and started
STEP: CAL: SPLIT TEST     <- step timer started
CUE: PRIMING — NO NEED TO WATCH YET
CUE: EYES ON DISP VALVE — FRONT ARRIVES T-11s
CUE: NOW                  <- the strobe moment
DONE 183s                 <- finished, measured duration
ABORTED                   <- (instead of DONE) the stop latch fired

```

The Step-8 stepwise toolkit (`b` → `u` → `m / M`)

Decoupled bubble-calibration protocol, every advance operator-gated:

1. `d a r` — the **clean regime**. Run it before every reading (`r` alone does not clean the six sample lines).
2. `b` — park the bubble with full production fidelity, then freeze it (line sealed; the front cannot creep). Read the front against the valve.
3. `u` (repeat) — advance ~13 µL per press until the first air tip exits into line 7. The printed total is the measured reagent-line volume.
4. `M` — the production split in one step; inspect the six air slugs. Or `m` six times — the same split one line at a time, resealing between slices so every intermediate state can be read.
5. `p` — the production purge; inspect the six wash edges at the needles.

See `KNOWN-ISSUES.md` for what these tests established on 2026-08-21 and what remains open.

3. Rules that protect the hardware

- **Never flash while a run is active** — uploading reboots the board mid-motion. Confirm `s` shows IDLE first.

- **Pump arming:** after any front-panel keypad use, press the pump's STOP key once; remote control only arms after STOP. The keypad locks out while the remote contact is closed.
- **Power sequence:** GIGA on **first**, pump power off **first**. An unpowered GIGA drives the pump's start input low — the pump runs.
- **Port 6 of the reagent valve must stay dry** (no tubing): routines draw air through it. **Port 1 of the sample and vacuum valves is never plumbed** — the software parks there and the run logic requires it open.